

20 General References

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21 Authors

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22 Date of Revision

12 February 2009.

Alpha Tocopherol

1 Nonproprietary Names

BP: RRR-Alpha-Tocopherol

JP: Tocopherol

PhEur: RRR- α -Tocopherol

USP: Vitamin E

See also Sections 3, 9, and 17.

2 Synonyms

Copherol F1300; ($\pm$)-3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-ol; E307; RRR- α -tocopherolum; synthetic alpha tocopherol; all-*rac*- α -tocopherol; *dl*- α -tocopherol; 5,7,8-trimethyltolcol.

3 Chemical Name and CAS Registry Number

($\pm$)-(2*RS*,4'*RS*,8'*RS*)-2,5,7,8-Tetramethyl-2-(4',8',12'-trimethyltridecyl)-6-chromanol [10191-41-0]

Note that alpha tocopherol has three chiral centers, giving rise to eight isomeric forms. The naturally occurring form is known as *d*-alpha tocopherol or (2*R*,4'*R*,8'*R*)-alpha-tocopherol. The synthetic form, *dl*-alpha tocopherol or simply alpha tocopherol, occurs as a racemic mixture containing equimolar quantities of all the isomers.

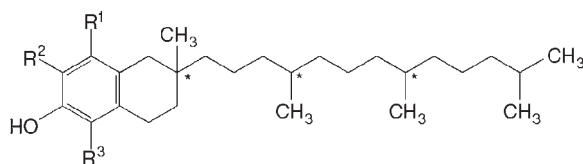
Similar considerations apply to beta, delta, and gamma tocopherol and tocopherol esters.

See Section 17 for further information.

4 Empirical Formula and Molecular Weight

C₂₉H₅₀O₂ 430.72

5 Structural Formula



Alpha tocopherol: R¹ = R² = R³ = CH₃

Beta tocopherol: R¹ = R³ = CH₃; R² = H

Delta tocopherol: R¹ = CH₃; R² = R³ = H

Gamma tocopherol: R¹ = R² = CH₃; R³ = H

* Indicates chiral centers.

6 Functional Category

Antioxidant; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Alpha tocopherol is primarily recognized as a source of vitamin E, and the commercially available materials and specifications reflect this purpose. While alpha tocopherol also exhibits antioxidant properties, the beta, delta, and gamma tocopherols are considered to be more effective as antioxidants.

Alpha-tocopherol is a highly lipophilic compound, and is an excellent solvent for many poorly soluble drugs.^(1–4) Of widespread regulatory acceptability, tocopherols are of value in oil- or fat-based pharmaceutical products and are normally used in the concentration range 0.001–0.05% v/v. There is frequently an optimum concentration; thus the autoxidation of linoleic acid and methyl linolenate is reduced at low concentrations of alpha tocopherol, and is accelerated by higher concentrations. Antioxidant effectiveness can be increased by the addition of oil-soluble synergists such as lecithin and ascorbyl palmitate.⁽⁴⁾

Alpha tocopherol may be used as an efficient plasticizer.⁽⁵⁾ It has been used in the development of deformable liposomes as topical formulations.⁽⁶⁾

d-Alpha-tocopherol has also been used as a non-ionic surfactant in oral and injectable formulations.⁽³⁾

8 Description

Alpha tocopherol is a natural product. The PhEur 6.0 describes alpha-tocopherol as a clear, colorless or yellowish-brown, viscous, oily liquid. See also Section 17.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for alpha tocopherol.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	—	+	—
Acidity	—	—	+
Optical rotation	—	+0.05° to +0.10°	+
Heavy metals	≤20 ppm	—	—
Related substances	—	+	—
Absorbance	+	—	—
at 292 nm	71.0–76.0	—	—
Refractive index	1.503–1.507	—	—
Specific gravity	0.947–0.955	—	—
Clarity and color of solution	+	—	—
Assay	96.0–102.0%	94.5–102.0%	96.0–102.0%

Note that the USP 32 describes vitamin E as comprising *d*- or *dl*-alpha tocopherol, *d*- or *dl*-alpha tocopheryl acetate, or *d*- or *dl*-alpha tocopheryl acid succinate. However, the PhEur 6.0 describes alpha tocopherol and alpha tocopheryl acetate in separate monographs.

The diversity of the tocopherols described in the various pharmacopeial monographs makes the comparison of specifications more complicated; see Section 17.

10 Typical Properties

Boiling point 235°C

Density 0.947–0.951 g/cm³

Flash point 240°C

Ignition point 340°C

Refractive index $n_D^{20} = 1.503\text{--}1.507$

Solubility Practically insoluble in water; freely soluble in acetone, ethanol, ether, and vegetable oils.

11 Stability and Storage Conditions

Tocopherols are oxidized slowly by atmospheric oxygen and rapidly by ferric and silver salts. Oxidation products include tocopheroxide, tocopherylquinone, and tocopherylhydroquinone, as well as dimers and trimers. Tocopherol esters are more stable to oxidation than the free tocopherols but are in consequence less effective antioxidants. See also Section 17.

Tocopherols should be stored under an inert gas, in an airtight container in a cool, dry place and protected from light.

12 Incompatibilities

Tocopherols are incompatible with peroxides and metal ions, especially iron, copper, and silver. Tocopherols may be absorbed into plastic.⁽⁷⁾

13 Method of Manufacture

Naturally occurring tocopherols are obtained by the extraction or molecular distillation of steam distillates of vegetable oils; for example, alpha tocopherol occurs in concentrations of 0.1–0.3% in corn, rapeseed, soybean, sunflower, and wheat germ oils.⁽⁸⁾ Beta and gamma tocopherol are usually found in natural sources along with alpha tocopherol. Racemic synthetic tocopherols may be prepared by the condensation of the appropriate methylated hydroquinone with racemic isophytol.⁽⁹⁾

14 Safety

Tocopherols (vitamin E) occur in many food substances that are consumed as part of the normal diet. The daily nutritional requirement has not been clearly defined but is estimated to be 3.0–20.0 mg. Absorption from the gastrointestinal tract is dependent upon normal pancreatic function and the presence of bile. Tocopherols are widely distributed throughout the body, with some ingested tocopherol metabolized in the liver; excretion of metabolites is via the urine or bile. Individuals with vitamin E deficiency are usually treated by oral administration of tocopherols, although intramuscular and intravenous administration may sometimes be used.

Tocopherols are well tolerated, although excessive oral intake may cause headache, fatigue, weakness, digestive disturbance, and nausea. Prolonged and intensive skin contact may lead to erythema and contact dermatitis.

The use of tocopherols as antioxidants in pharmaceuticals and food products is unlikely to pose any hazard to human health since the daily intake from such uses is small compared with the intake of naturally occurring tocopherols in the diet.

The WHO has set an acceptable daily intake of tocopherol used as an antioxidant at 0.15–2.0 mg/kg body-weight.⁽¹⁰⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Gloves and eye protection are recommended.

16 Regulatory Status

GRAS listed. Accepted in Europe as a food additive. Included in the FDA Inactive Ingredients Database (IV injections, powder, lyophilized powder for liposomal suspension; oral capsules, tablets, and topical preparations). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

17 Related Substances

d-Alpha tocopherol; *d*-alpha tocopheryl acetate; *dl*-alpha tocopheryl acetate; *d*-alpha tocopheryl acid succinate; *dl*-alpha tocopheryl acid succinate; beta tocopherol; delta tocopherol; gamma tocopherol; tocopherols excipient.

d-Alpha tocopherol

Empirical formula C₂₉H₅₀O₂

Molecular weight 430.72

CAS number [59-02-9]

Synonyms Natural alpha tocopherol; (+)-(2*R*,4'*R*,8'*R*)-2,5,7,8-tetramethyl-2-(4',8',12'-trimethyltridecyl)-6-chroman-ol; *d*-α-tocopherol; vitamin E.

Appearance A practically odorless, clear, yellow, or greenish-yellow viscous oil.

Melting point 2.5–3.5°C

Solubility Practically insoluble in water; soluble in ethanol (95%). Miscible with acetone, chloroform, ether, and vegetable oils.

Specific gravity 0.95

Comments *d*-Alpha tocopherol is the naturally occurring form of alpha tocopherol.

d-Alpha tocopheryl acetate

Empirical formula C₃₁H₅₂O₃

Molecular weight 472.73

CAS number [58-95-7]

Synonyms (+)-(2*R*,4'*R*,8'*R*)-2,5,7,8-Tetramethyl-2-(4',8',12'-trimethyltridecyl)-6-chromanyl acetate; *d*-α-tocopheryl acetate; vitamin E.

Appearance A practically odorless, clear, yellow, or greenish-yellow colored viscous oil that may solidify in the cold.

Melting point 28°C

Solubility Practically insoluble in water; soluble in ethanol (95%). Miscible with acetone, chloroform, ether, and vegetable oils.

Specific rotation $[\alpha]_D^{25} = +0.25^\circ$ (10% w/v solution in chloroform)

Comments Unstable to alkalis.

dl-Alpha tocopheryl acetate

Empirical formula C₃₁H₅₂O₃

Molecular weight 472.73

CAS number [7695-91-2]

Synonyms (±)-3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2*H*-1-benzopyran-6-ol acetate; (±)-(2*RS*,4'*RS*,8'*RS*)-2,5,7,8-tetramethyl-2-(4',8',12'-trimethyltridecyl)-6-chromanyl acetate; (±)-α-tocopherol acetate; α-tocopherol acetate; all-*rac*-α-tocopheryl acetate; *dl*-α-tocopheryl acetate; vitamin E.

Appearance A practically odorless, clear, yellow, or greenish-yellow viscous oil.

Density 0.953 g/cm³

Melting point –27.5°C

Refractive index $n_D^{20} = 1.4950\text{--}1.4972$

Solubility Practically insoluble in water; freely soluble in acetone, chloroform, ethanol, ether, and vegetable oils; soluble in ethanol (95%).

Comments Unstable to alkali. However, unlike alpha tocopherol, the acetate is much less susceptible to the effects of air, light, or

ultraviolet light. Alpha tocopherol acetate concentrate, a powdered form of alpha tocopherol acetate, is described in the PhEur 6.0. The concentrate may be prepared by either dispersing alpha tocopherol acetate in a suitable carrier such as acacia or gelatin, or by adsorbing alpha tocopherol acetate on silicic acid.

d-Alpha tocopheryl acid succinate

Empirical formula $C_{33}H_{54}O_5$

Molecular weight 530.8

CAS number [4345-03-3]

Synonyms (+)- α -Tocopherol hydrogen succinate; *d*- α -tocopheryl acid succinate; vitamin E.

Appearance A practically odorless white powder.

Melting point 76–77°C

Solubility Practically insoluble in water; slightly soluble in alkaline solutions; soluble in acetone, ethanol (95%), ether, and vegetable oils; very soluble in chloroform.

Comments Unstable to alkalis.

dl-Alpha tocopheryl acid succinate

Empirical formula $C_{33}H_{54}O_5$

Molecular weight 530.8

CAS number [17407-37-3]

Synonyms ($\pm$)- α -Tocopherol hydrogen succinate; *dl*- α -tocopheryl acid succinate; *dl*- α -tocopherol succinate; vitamin E.

Appearance A practically odorless, white crystalline powder.

Solubility Practically insoluble in water; slightly soluble in alkaline solutions; soluble in acetone, ethanol (95%), ether, and vegetable oils; very soluble in chloroform.

Comments Unstable to alkalis.

Beta tocopherol

Empirical formula $C_{28}H_{48}O_2$

Molecular weight 416.66

CAS number [148-03-8]

Synonyms Cumotocopherol; ($\pm$)-3,4-dihydro-2,5,8-trimethyl-2-(4,8,12-trimethyltridecyl)-2H-1- β -benzopyran-6-ol; 5,8-dimethyltolcol; neotocopherol; *dl*- β -tocopherol; vitamin E; *p*-xylotocopherol.

Appearance A pale yellow-colored viscous oil.

Solubility Practically insoluble in water; freely soluble in acetone, chloroform, ethanol (95%), ether, and vegetable oils.

Specific rotation $[\alpha]_D^{20} = +6.37^\circ$

Comments Less active biologically than alpha tocopherol. Obtained along with alpha tocopherol and gamma tocopherol from natural sources. Beta tocopherol is very stable to heat and alkalis and is slowly oxidized by atmospheric oxygen.

Delta tocopherol

Empirical formula $C_{27}H_{46}O_2$

Molecular weight 402.64

CAS number [119-13-1]

Synonyms ($\pm$)-3,4-Dihydro-2,8-dimethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-ol; E309; 8-methyltolcol; *dl*- δ -tocopherol; vitamin E.

Appearance A pale yellow-colored viscous oil.

Solubility Practically insoluble in water; freely soluble in acetone, chloroform, ethanol (95%), ether, and vegetable oils.

Comments Occurs naturally as 30% of the tocopherol content of soybean oil. Delta tocopherol is said to be the most potent antioxidant of the tocopherols.

Gamma tocopherol

Empirical formula $C_{28}H_{48}O_2$

Molecular weight 416.66

CAS number [7616-22-0]

Synonyms ($\pm$)-3,4-Dihydro-2,7,8-trimethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-ol; 7,8-dimethyltolcol; E308; *dl*- γ -tocopherol; vitamin E; *o*-xylotocopherol.

Appearance A pale yellow-colored viscous oil.

Melting point -30°C

Solubility Practically insoluble in water; freely soluble in acetone, chloroform, ethanol (95%), ether, and vegetable oils.

Specific rotation $[\alpha]_D^{20} = -2.4^\circ$ (in ethanol (95%))

Comments Occurs in natural sources along with alpha and beta tocopherol. Gamma tocopherol is biologically less active than alpha tocopherol. Very stable to heat and alkalis; slowly oxidized by atmospheric oxygen and gradually darkens on exposure to light.

Tocopherols excipient

Synonyms *Embanox tocopherol*.

Appearance A pale yellow-colored viscous oil.

Comments Tocopherols excipient is described in the USP32–NF27 as a vegetable oil solution containing not less than 50.0% of total tocopherols, of which not less than 80.0% consists of varying amounts of beta, delta, and gamma tocopherols.

18 Comments

Note that most commercially available tocopherols are used as sources of vitamin E, rather than as antioxidants in pharmaceutical formulations.

Various mixtures of tocopherols, and mixtures of tocopherols with other excipients, are commercially available, and individual manufacturers should be consulted for specific information on their products.

Molecularly imprinted polymers for use in the controlled release of alpha tocopherol in gastrointestinal simulating fluids have been investigated.⁽¹⁾

The EINECS number for α -tocopherol is 215-798-8. The EINECS number for *d*- α -tocopherol is 200-412-2; and the EINECS number for *dl*- α -tocopherol is 233-466-0. The PubChem Compound ID (CID) for alpha tocopherol includes 14985 and 1548900.

19 Specific References

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20 General References

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21 Author

ME Quinn.

22 Date of Revision

28 January 2009.



Aluminum Hydroxide Adjuvant

1 Nonproprietary Names

PhEur: Aluminium Hydroxide, Hydrated, for Adsorption

2 Synonyms

Alhydrogel; aluminii hydroxidum hydricum ad adsorptionem; aluminium hydroxide adjuvant; aluminium oxyhydroxide; poorly crystalline boehmite; pseudoboehmite; *Rehydrigel*.

3 Chemical Name and CAS Registry Number

Aluminum oxyhydroxide [21645-51-2]

4 Empirical Formula and Molecular Weight

AlO(OH) 59.99

5 Structural Formula

Structural hydroxyl groups form hydrogen bonds between AlO(OH) octahedral sheets, where hydroxyl groups are exposed at the surface. The surface hydroxyl groups produce a pH-dependent surface charge by accepting a proton to produce a positive site, or donating a proton to produce a negative site. The pH-dependent surface charge is characterized by the point of zero charge, which is equivalent to the isoelectric point in protein chemistry. The surface hydroxyl groups may also undergo ligand exchange with fluoride, phosphate, carbonate, sulfate, or borate anions.

6 Functional Category

Adsorbent; vaccine adjuvant.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum hydroxide adjuvant is used in parenteral human and veterinary vaccines.⁽¹⁾ It activates Th2 immune responses, including IgG and IgE antibody responses. It is also used for the isolation of certain serum components such as blood clotting factors.⁽²⁾

8 Description

Aluminum hydroxide adjuvant is a white hydrogel that sediments slowly and forms a clear supernatant.

9 Pharmacopeial Specifications

See Table I. Note that the USP 32 includes a monograph for aluminum hydroxide gel, which is a form of aluminum hydroxide that is used as an antacid, in which there is a partial substitution of carbonate for hydroxide.

See Section 17.

Table I: Pharmacopeial specifications for aluminum hydroxide adjuvant.

Test	PhEur 6.1
Identification	+
Characters	+
Solution	+
pH	5.5–8.5
Adsorption power	+
Sedimentation	+
Chlorides	≤ 0.33%
Nitrates	≤ 100 ppm
Sulfates	≤ 0.5%
Ammonium	≤ 50 ppm
Arsenic	≤ 1 ppm
Iron	≤ 5 ppm
Heavy metals	≤ 20 ppm
Bacterial endotoxins	+
Assay	90.0–110.0%

10 Typical Properties

Acidity/alkalinity pH = 5.5–8.5

Particle size distribution Primary particles are fibrous with average dimensions of $4.5 \times 2.2 \times 10$ nm. The primary particles form aggregates of 1–10 μ m.

Point of zero charge pH = 11.4

Protein binding capacity >0.5 mg BSA/mg equivalent Al₂O₃

Solubility Soluble in alkali hydroxides and mineral acids. Heat may be required to dissolve the aluminum hydroxide adjuvant.

Specific surface area 500 m²/g.⁽³⁾

X-ray diffractogram Exhibits characteristic x-ray diffraction pattern having diffraction bands at 6.46, 3.18, 2.35, 1.86, 1.44 and 1.31 Å.

11 Stability and Storage Conditions

Aluminum hydroxide adjuvant is stable for at least 2 years when stored at 4–30°C in well-sealed inert containers. It must not be allowed to freeze as the hydrated colloid structure will be irreversibly damaged.

12 Incompatibilities

When exposed to phosphate, carbonate, sulfate, or borate anions, the point of zero charge for aluminum hydroxide adjuvant decreases.

13 Method of Manufacture

Aluminum hydroxide adjuvant is prepared by the precipitation of a soluble aluminum salt by an alkali hydroxide, or the precipitation of an alkali aluminate by acid.

14 Safety

Aluminum hydroxide adjuvant is intended for use in parenteral vaccines and is generally regarded as nontoxic. It may cause mild irritation, dryness, and dermatitis on skin contact. On eye contact, aluminum hydroxide adjuvant may also cause redness, conjunctivitis, and short-term mild irritation. Ingestion of large amounts may cause gastrointestinal irritation with nausea, vomiting, and constipation. Inhalation of the dried product may cause respiratory irritation and cough. Type I hypersensitivity reactions following parenteral administration have been reported.⁽⁴⁾